

# MC Vertex Reweighting for PPG12:

## A Data-Driven Iterative Method on the Truth Vertex

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### Abstract

The PPG12 efficiency pipeline must reweight MC to match the measured  $p+p$  luminous region, which is very different between the two crossing angles ( $\sigma \simeq 53$  cm at 0 mrad vs  $\sigma \simeq 22$  cm at 1.5 mrad, Piwinski narrowed). The production reco-level weight  $f(z_r) = D(z_r)/M(z_r)$  closes the reco vertex distribution exactly, but it leaves the *individual* truth-vertex widths of double-interaction MC events at a geometric “half narrow, half wide” compromise ( $\sigma \simeq 47$  cm instead of 22 cm), which in turn biases the truth fiducial cut used by the efficiency denominator. The resulting cross-section bias at 1.5 mrad is +1.4% — comparable to other PPG12 systematics and well within reach of a correctable method. This note presents the nominal fix: an **iterative data-driven reweight**  $w(z)$  applied per truth vertex ( $w(z_h)$  for single interactions,  $w(z_h)w(z_{mb})$  for doubles). The method uses no physics prior for the target shape — only the empirical width of the data reco distribution as the seed Gaussian, and the 99-th percentile of  $|z_r^{\text{data}}|$  as the auto-derived closure window. Applied to the photon10 mix, the method converges cleanly at 0 mrad ( $\max|D/M-1| < 1\%$  in 8 iterations) and reduces  $\chi^2$  in the closure window by a factor  $\sim 3100$  at 1.5 mrad, closing the bulk of the window to the 1–2% level inside  $|z_r| < 30$  cm and reaching  $\sim 11\%$  at the closure-window edge. The code is a standalone Python package in `efficiencytool/truth_vertex_reweight/`; the per-period `reweight.root` outputs are ready to be consumed by `RecoEffCalculator_TTreeReader.C`.

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# 1 Problem and motivation

PPG12 extracts the isolated prompt-photon cross section from jet-dominated  $p+p$  triggered data. The cross section is proportional to the measured yield divided by the MC-estimated signal efficiency  $\epsilon$ . Both the numerator (reco clusters in data, passing the selection) and the denominator (MC truth photons in the fiducial window) inherit the MC vertex distribution — the MC prior on the vertex  $z$ -coordinate enters via (i) cluster kinematics reconstructed relative to the reco vertex  $z_r$ , (ii) the  $|z_r| < \text{vertex\_cut}$  cut applied in data and MC alike, and (iii) the truth fiducial cut  $|z_h| < \text{vertex\_cut}$  used for the efficiency denominator. *Any* data/MC mismatch in the vertex  $z$  distribution (reco or truth) therefore feeds directly into  $\epsilon$  and the extracted cross section.

**Notation.** Throughout this note,  $z_r$  denotes the MBD-reconstructed vertex;  $z_h \equiv z_{t,1}$  denotes the primary (hard-scatter) truth vertex; and  $z_{mb} \equiv z_{t,2}$  denotes the secondary (minimum-bias) truth vertex, present only in double-interaction events. For double-interaction MC,  $z_r \approx (z_h + z_{mb})/2$  up to MBD resolution. Sections 1–2 occasionally use the  $(z_{t,1}, z_{t,2})$  pair notation when the algebraic decomposition into symmetric/antisymmetric coordinates is needed; elsewhere we use the  $(z_h, z_{mb})$  notation exclusively.

**The two periods are very different.** At the 0 mrad crossing angle the beams overlap along their full length and the luminous region is  $\sigma_z \simeq 53$  cm. At 1.5 mrad the beams cross at an angle, the beam-beam overlap is geometrically shortened by the Piwinski factor, and the luminous region collapses to  $\sigma_z \simeq 22$  cm. The MC samples used in PPG12 were generated with the full  $\sim 65$  cm beam profile regardless of period; the reweight has to correct this downward by nearly a factor of three at 1.5 mrad. This is a very large shape correction and any shortcomings of the reweight method are amplified at 1.5 mrad.

**The production method.** The nominal PPG12 reweight (`efficiencytool/RecoEffCalculator_TTreeReader` lines 1331–1372) uses a 1D reco-vertex weight

$$f(z_r) = \frac{D(z_r)}{M_{\text{reco}}(z_r)},$$

where  $D(z_r)$  is the data reco-vertex shape and  $M_{\text{reco}}(z_r)$  is the MC reco-vertex shape, both area-normalized. Event-by-event, each MC event is multiplied by  $f(z_r^{\text{event}})$  using its reconstructed vertex. By construction,  $M'_{\text{reco}}(z_r) = f(z_r) \cdot M_{\text{reco}}(z_r) = D(z_r)$  bin by bin: the reweighted MC reco-vertex distribution is identical to data.

**Why the production method is not enough.**  $f(z_r)$  closes the reco distribution exactly, but it does not close the individual truth-vertex distributions of double-interaction MC events, and it is those truth vertices that enter the efficiency denominator via the truth fiducial cut. The structural reason is a clean algebraic one: decompose the pair  $(z_{t,1}, z_{t,2}) = (z_h, z_{mb})$  of truth vertices for a double-interaction event into a symmetric coordinate  $\bar{z} \equiv (z_{t,1} + z_{t,2})/2$  and an antisymmetric coordinate  $\varepsilon \equiv (z_{t,1} - z_{t,2})/2$ . Under the independent Gaussian sampling of the raw MC,  $\bar{z}$  and  $\varepsilon$  are statistically independent zero-mean Gaussians with  $\sigma_{\bar{z}} = \sigma_{\varepsilon} = \sigma_{\text{MC}}/\sqrt{2}$ . The reco vertex is  $z_r \approx \bar{z}$  (up to MBD resolution), so the production weight  $f(z_r)$  depends *only* on  $\bar{z}$  and is flat in  $\varepsilon$ . After reweighting,  $\bar{z}$  matches the target Gaussian  $\mathcal{N}(0, \sigma_{\text{lum}}^2/2)$  by construction but  $\varepsilon$  is unchanged. The primary truth vertex  $z_h = \bar{z} + \varepsilon$  is therefore a sum of a narrowed and an un-narrowed Gaussian,

$$\sigma^2(z_h) = \frac{1}{2} \sigma_{\text{lum}}^2 + \frac{1}{2} \sigma_{\text{MC}}^2 = \frac{1}{2} (22.3^2 + 62.7^2) \simeq (47.0 \text{ cm})^2. \quad (1)$$

$f(z_r)$  cannot touch  $\varepsilon$ , so it cannot narrow  $z_h$  and  $z_{mb}$  individually below this “half-narrow, half-wide” compromise. Any principled fix must therefore act on the truth vertices directly, i.e. it must factorize as  $w(z_h)w(z_{mb})$  rather than depending only on the reco combination  $z_r$ . The nominal method of Section 3 does exactly this, with a data-driven  $w(z)$  iterated to closure on the mixed-MC reco distribution.

## 2 Cross-section impact at 1.5 mrad

The efficiency denominator in `RecoEffCalculator_TTreeReader.C` uses both a reco-vertex cut  $|z_r| < \text{vertex\_cut}$  and a truth-vertex cut  $|z_h| < \text{vertex\_cut}$ , with `vertex_cut` = 60 cm in `config_bdt_nom.yaml`. The reco cut is handled correctly by  $f(z_r)$  because the reco distribution is closed to data exactly. The truth cut is *not* handled correctly because, as Eq. (1) shows,  $f(z_r)$  leaves  $\sigma(z_h) \simeq 47$  cm instead of the target 22 cm for double-interaction MC.

**Quantitative estimate.** For a Gaussian truth distribution of width  $\sigma$ , the fraction in  $|z_h| < 60$  cm is  $\text{erf}(60/(\sigma\sqrt{2}))$ . Plugging in:

- $\sigma = 47$  cm (what nominal  $f(z_r)$  gives):  $\text{erf}(60/(47\sqrt{2})) \simeq 0.80$
- $\sigma = 22$  cm (correct 1.5 mrad):  $\text{erf}(60/(22\sqrt{2})) \simeq 0.99$

The double-interaction MC loses  $\sim 19$  percentage points of truth fiducial acceptance that it shouldn't. The mixed MC blend at 1.5 mrad is  $0.921\text{single} + 0.079\text{double}$ , so the bias on the integrated efficiency denominator is

$$0.079 \times 0.19 \simeq 1.4\% \quad \text{at 1.5 mrad.}$$

This translates directly to a  $+1.4\%$  bias on the extracted cross section at 1.5 mrad (the efficiency denominator is too small, so the quoted cross section is too large). At 0 mrad the analogous bias is  $\sim 0.3\%$  because the MC beam profile is already close to the (wider) 0 mrad luminous region. At the percent level the current PPG12 production is biased at 1.5 mrad because  $f(z_r)$  does not correct the individual truth vertices  $z_h$  and  $z_{mb}$  of double-interaction events. The bias is comparable to other PPG12 systematics and is easily fixable with a truth-level reweight.

## 3 Method: data-driven iterative reweight

The method derives a truth-vertex weight function  $w(z)$  — evaluated at  $z_h$  for single-interaction MC and at both  $z_h$  and  $z_{mb}$  for double-interaction MC — directly from data, with no Gaussian assumption on the luminous-region shape, by iterating the mixed-MC reco distribution to closure against the data reco distribution.

**Forward model.** Let  $f_d$  be the cluster-weighted double-interaction fraction in a given crossing angle (0.079 at 1.5 mrad, 0.224 at 0 mrad) and  $f_s = 1 - f_d$  the single-interaction fraction. The mixed MC reco distribution under a candidate truth-vertex weight  $w(z)$  is

$$M(z_r) = f_s \sum_{e \in T_s} w(z_h^e) \delta(z_r - z_r^e) + f_d \sum_{e \in T_d} w(z_h^e) w(z_{mb}^e) \delta(z_r - z_r^e), \quad (2)$$

evaluated event-by-event over the single-interaction sample  $T_s$  and the double-interaction sample  $T_d$ . For double-interaction events the weight enters *twice*, once for the primary truth vertex  $z_h$  and once for the secondary truth vertex  $z_{mb}$ , reflecting the statistical independence of the two truth vertices under the generator. This is the crucial structural choice: it gives the iteration access to the antisymmetric coordinate  $\varepsilon$  of Section 1 that the production reco-only weight  $f(z_r)$  cannot touch.

**Binning and closure window.** Both periods use the same binning: 80 bins over  $[-200, 200]$  cm (bin width 5 cm), no overflow clipping, and identical bin edges for the  $w(z)$  weight function and the reco-vertex histograms. The iterative closure window is auto-derived per period as the 99-th percentile of  $|z_r^{\text{data}}|$ :

- 1.5 mrad:  $z_{\text{cut}} = 83$  cm (34 bins inside), from  $|z_r^{\text{data}}|_{99\%} = 82.0$  cm.
- 0 mrad:  $z_{\text{cut}} = 144$  cm (58 bins inside), from  $|z_r^{\text{data}}|_{99\%} = 143.9$  cm.

The window is purely data-driven and requires no manual tuning. Outside it, the statistics in data drop below the level where the iteration could meaningfully constrain  $w(z)$ .

**Seed.** The iteration starts from a Gaussian seed  $w_0(z) = \mathcal{N}(0, \sigma_{\text{seed}}^2) / \mathcal{N}(0, \sigma_{\text{gen}}^2)$ , with  $\sigma_{\text{gen}}$  measured from the photon10 primary truth vertex  $z_h$  (`vertexz_truth` branch) ( $\sigma_{\text{gen}} = 64.997 \pm 0.015$  cm, intrinsic to the generator and identical between periods, since the same single MC is loaded for both) and  $\sigma_{\text{seed}} = \text{std}(z_r^{\text{data}})$  measured directly from the data reco-vertex distribution in the relevant run range (21.48 cm at 1.5 mrad, 54.32 cm at 0 mrad). Note that  $\sigma_{\text{seed}}$  requires *no* physics input — no Piwinski narrowing factor, no luminous-region calculation. It is just the empirical width of the reco distribution we are trying to match.

**Iterative update.** At iteration  $i$ , the residual is  $r_i(z_r) = D(z_r)/M_i(z_r)$ , where  $D(z_r)$  is the data reco-vertex histogram and  $M_i(z_r)$  is the mixed MC reco-vertex histogram from Eq. (2) under the current  $w_i$ . The update is Lucy–Richardson-like, multiplicative, and damped:

$$w_{i+1}(z) = w_i(z) \cdot [\text{smooth}(r_i(z))]^\alpha, \quad \alpha = 0.5, \quad (3)$$

where the residual  $r_i$  evaluated at reco bin  $z_r$  is applied to the weight at the matching truth bin  $z = z_r$  (using the shared binning). After each update,  $w$  is renormalized so that  $\langle w(z_h) \rangle$  averaged over the single-MC truth distribution equals unity, preserving the physics double-interaction fraction. Damping ( $\alpha = 0.5$ ) prevents over-correction; a 1-bin Gaussian smoothing on the residual prevents bin-by-bin oscillation; the update factor is clipped to  $[0.5, 2]$  per iteration to bound jumps; and the endpoints of the update region are edge-clamped to their nearest interior values to prevent runaway behavior at the boundary. The update is applied over the full 80-bin binning and the final  $w(z)$  is data-driven everywhere, with no post-iteration snap to a seed Gaussian. (An earlier version of the method snapped  $w(z)$  to the seed Gaussian outside the closure window; a systematic study with `compare_snap_strategies.py` showed that removing the snap improves 0 mrad closure by a factor  $\sim 1.3$  in  $\chi^2$  and  $\sim 3$  in  $\max|r-1|$  while leaving 1.5 mrad essentially unchanged.) Convergence is tested as  $\max|r_i(z)-1| < 0.01$  inside the closure window, with a hard cap of 30 iterations.

**Implementation.** The method is a standalone Python tool in `efficiencytool/truth_vertex_reweight/`. The mixed-MC histogram is heavy to rebuild ( $\sim 22$  M events at 1.5 mrad,  $\sim 48$  M at 0 mrad), so per-sample arrays are cached to `output/cache/*.npz` keyed on file mtime; each iteration runs in seconds per iteration on a single core. A short file list is in Appendix D. Per-period outputs live in `output/{1p5mrad,0mrad}/` and include `reweight.root` (histograms `h_w_iterative`, `h_w_seed`, `h_M_iterative`) and `fit_result.yaml`.

## 4 Per-period results

Both periods are fit on the photon10 / photon10\_double mix only at this stage; the full 4-sample blend including jet12/jet12\_double is a follow-up (Section 6). Table 1 collects the headline numbers from both fits.

| quantity                                                    | 1.5 mrad           | 0 mrad        |
|-------------------------------------------------------------|--------------------|---------------|
| run range                                                   | 51274–54000        | 47289–51274   |
| luminosity [pb <sup>-1</sup> ]                              | 16.8588            | 32.6574       |
| $f_{\text{double}}$ (cluster-weighted)                      | 0.079              | 0.224         |
| data $ z_r $ 99% percentile [cm]                            | 82.0               | 143.9         |
| closure window $z_{\text{cut}}$ [cm]                        | 83                 | 144           |
| bins inside closure window                                  | 34                 | 58            |
| $\sigma_{\text{seed}} = \text{std}(z_r^{\text{data}})$ [cm] | 21.48              | 54.32         |
| $\sigma_{\text{gen}}$ from MC $z_h$ (generator) [cm]        | $64.997 \pm 0.015$ |               |
| initial $\chi^2$ (in window)                                | 1,419,454          | 26,589        |
| final $\chi^2$ (in window)                                  | 453                | 158           |
| $\chi^2$ reduction factor                                   | $3132\times$       | $168\times$   |
| final $\chi^2/\text{ndf}$ (in window)                       | 13.7 (ndf 33)      | 2.78 (ndf 57) |
| initial $\max r-1 $                                         | 0.635              | 0.079         |
| final $\max r-1 $ (in window)                               | 0.111              | 0.008         |
| iterations used                                             | 30 (cap hit)       | 8             |
| converged to 0.01 tolerance                                 | <b>no</b>          | <b>yes</b>    |
| (context) parametric Gaussian $\chi^2/\text{ndf}$           | 3237               | 454           |

Table 1: Iterative data-driven reweight, photon10 mix only, both crossing angles, with the post-iteration snap to seed removed (Section 3). The 0 mrad fit converges to the 1 % per-bin tolerance in 8 iterations (final  $\max|r-1| \simeq 0.8\%$ ). The 1.5 mrad fit reduces  $\chi^2$  in the closure window by a factor  $\sim 3100$  in 30 iterations and stalls at  $\max|r-1| \simeq 11\%$ , dominated by bins at the closure-window edge  $|z_r| \simeq 83$  cm. Inside the bulk  $|z_r| < 30$  cm the per-bin closure is at the 1–2 % level (see Figure 4). The last row reports the parametric Gaussian  $\chi^2/\text{ndf}$  (best-fit  $\sigma_{\text{tgt}}$  from Stage-1) against the data; the iterative method improves on this by a factor of  $\sim 236$  at 1.5 mrad and  $\sim 163$  at 0 mrad.

**Hero summary.** Figure 1 is the single-figure summary: data reco vs reweighted MC reco overlaid for both periods, residuals, the recovered  $w(z)$ , and the headline numbers. The 0 mrad mix-MC matches data to  $\lesssim 1\%$  per bin throughout the closure window; the 1.5 mrad mix matches to  $\sim 1\text{--}2\%$  in the bulk  $|z_r| < 30$  cm, degrades to  $\sim 6\text{--}8\%$  around  $|z_r| \simeq 60$  cm, and reaches  $\sim 11\%$  at the closure-window edge  $|z_r| \simeq 83$  cm.

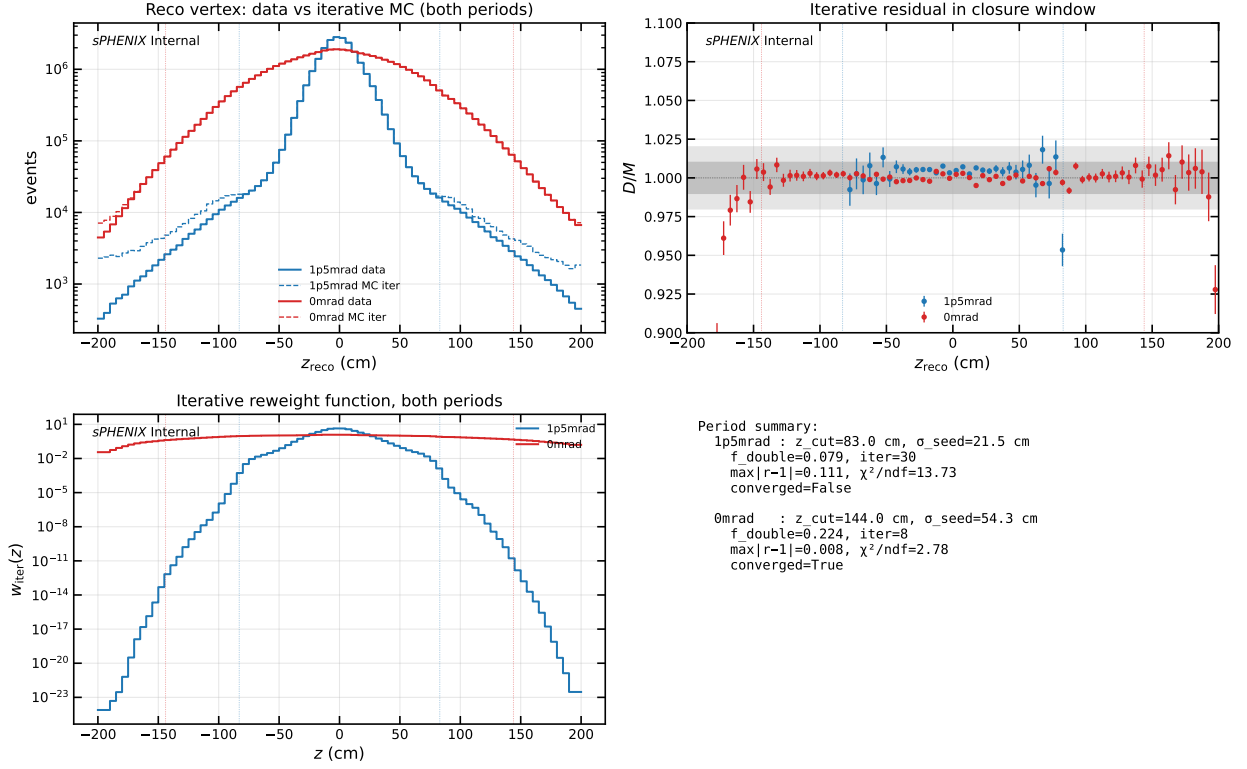


Figure 1: Iterative data-driven reweight summary, both crossing angles ( $2 \times 2$  grid plus summary text). Top-left: data reco  $z_r$  (solid) vs mixed MC reco  $z_r$  after iterative reweight (dashed), both periods overlaid (log scale). Top-right: residual  $r(z_r) = D(z_r)/M_{\text{iter}}(z_r)$  inside each period’s closure window (vertical dashed lines). Bottom-left: recovered truth-vertex weight  $w(z)$  for both periods (log scale). Bottom-right: summary statistics. The 0 mrad case is essentially closed; the 1.5 mrad case is closed in the bulk and stalls at the closure-window edge (see Section 5).

**Truth vertex distribution comparison.** Figure 2 overlays the primary truth vertex  $z_h$  distribution for the mixed MC under three weighting methods: no reweight (raw MC,  $\sigma \simeq 64$  cm), the production reco-level weight  $f(z_r)$ , and the new iterative truth-level weight  $w(z_h)$ . At 1.5 mrad the production method leaves the mixed-MC primary truth vertex  $z_h$  at  $\sigma \simeq 27$  cm — narrower than the raw MC but still far wider than the true luminous-region width ( $\sigma_{\text{lum}} \simeq 22$  cm), because  $f(z_r)$  cannot narrow  $z_h$  and  $z_{\text{mb}}$  individually for double-interaction events below the half-narrow/half-wide compromise (Section 1). The iterative method  $w(z_h)$  narrows the  $z_h$  distribution to  $\sigma \simeq 16$  cm, which is narrower than the data reco width  $\sigma_{\text{seed}} = 21.5$  cm precisely as expected, since each single-interaction truth vertex must be narrower than the observed reco width (the MBD vertex finder adds resolution smearing). At 0 mrad, where the correction is small, both methods converge to the same  $\sigma \simeq 54\text{--}57$  cm.

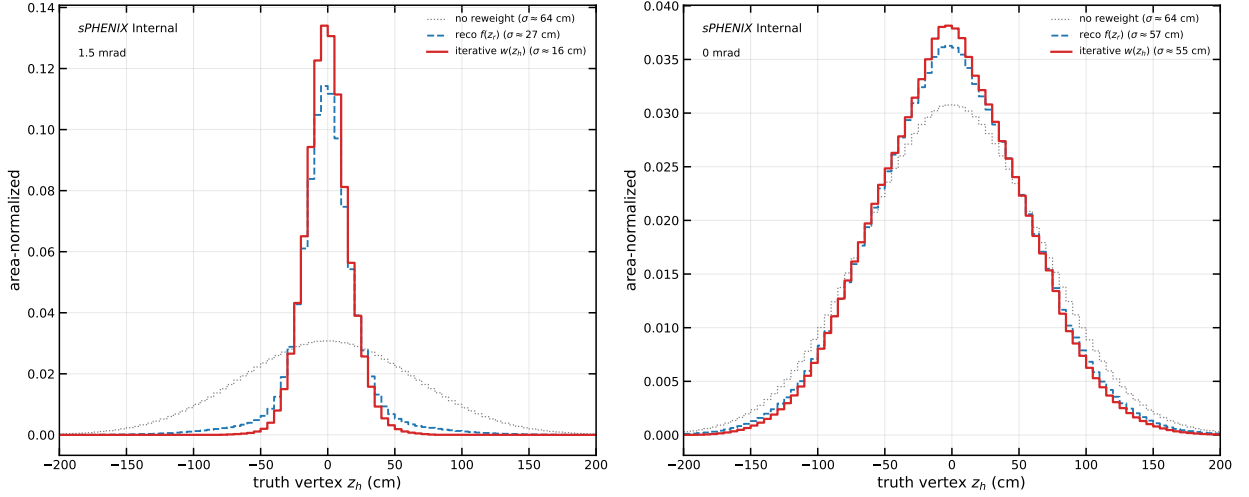


Figure 2: Truth vertex  $z_h$  distribution for the mixed photon10 MC under three reweighting methods, per period. Left: 1.5 mrad, where the reco-level  $f(z_r)$  (blue dashed) fails to narrow the truth distribution sufficiently, while the iterative  $w(z_h)$  (red) narrows it well past the data reco width (the truth must be narrower than the reco due to MBD resolution smearing). Right: 0 mrad, where both methods agree because the MC beam profile is already close to the data luminous region.

**0 mrad: clean convergence.** Figure 3 shows the 6-panel diagnostic for 0 mrad. The seed Gaussian with  $\sigma_{\text{seed}} \simeq 54$  cm already captures most of the shape (initial  $\max|r - 1| = 8\%$ , initial  $\chi^2/\text{ndf} \simeq 467$ ); the parametric best-fit Gaussian at  $\sigma_{\text{tgt}} \simeq 54.6$  cm is essentially the same ( $\chi^2/\text{ndf} \simeq 454$ ) because the data is nearly Gaussian. The iteration cleans up the residual over 8 updates and drops below the 1% per-bin tolerance inside the closure window (final  $\max|r - 1| \simeq 0.8\%$ ,  $\chi^2/\text{ndf} \simeq 2.78$ ). The recovered  $w(z)$  is smooth and broadly Gaussian, with a small enhancement of the central core relative to the seed, consistent with the data luminous region being slightly more peaked than a single Gaussian.

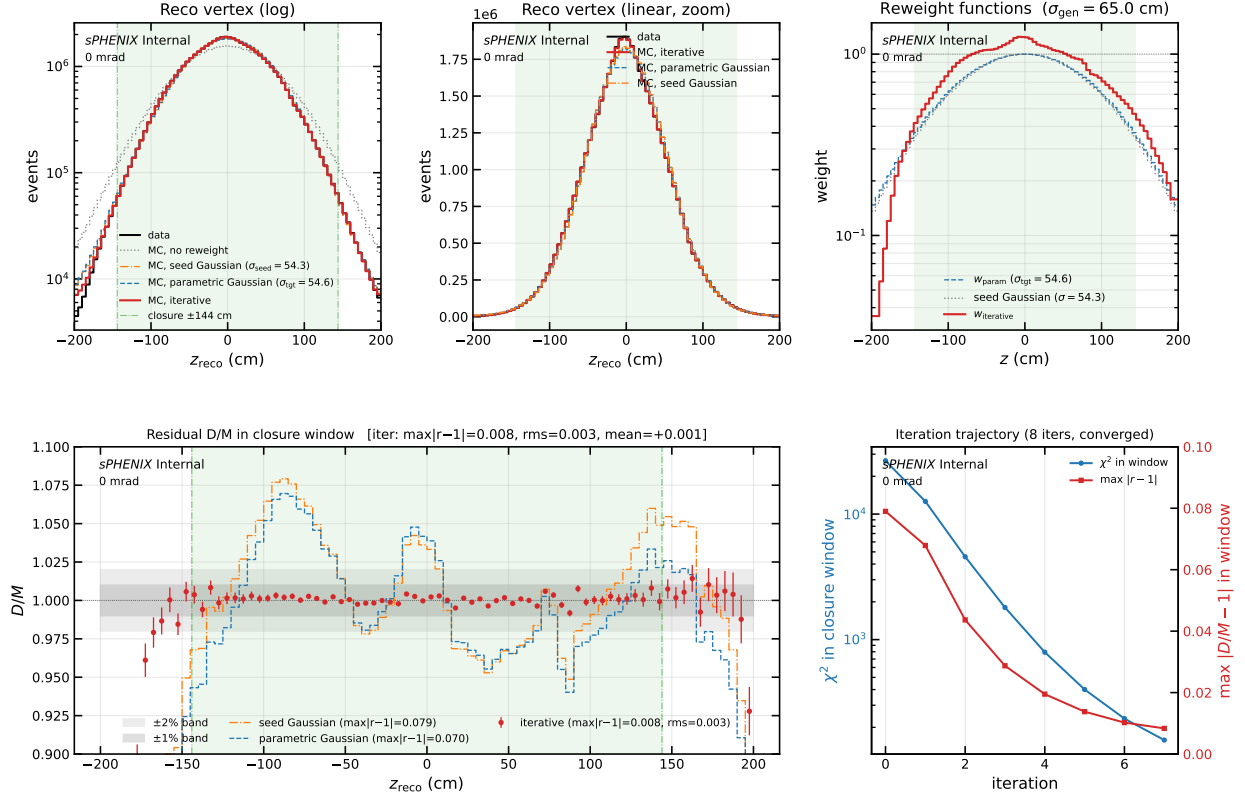


Figure 3: 0 mrad iterative closure (6 panels). Top-left/centre: data reco vs mixed MC reco under no reweight (gray dotted), the seed Gaussian  $w(z) \propto \mathcal{N}(0, \sigma_{\text{seed}}^2) / \mathcal{N}(0, \sigma_{\text{gen}}^2)$  with  $\sigma_{\text{seed}} = \text{std}(z_r^{\text{data}})$  (orange dash-dot), the parametric best-fit Gaussian with  $\sigma_{\text{tgt}}$  from the  $\chi^2$  scan (blue dashed), and the iterative reweight (red). Top-right: the corresponding reweight functions  $w(z)$  (seed, parametric, iterative). Bottom-left: residual  $D/M$  in the closure window with Poisson error bars on the iterative points and  $\pm 1\% / \pm 2\%$  reference bands. Bottom-right: per-iteration  $\chi^2$  and  $\max |r-1|$  trajectory. At 0 mrad the seed and parametric Gaussians are nearly identical ( $\sigma_{\text{seed}} \simeq \sigma_{\text{tgt}} \simeq 54$  cm) because the data reco distribution is already close to a single Gaussian; the iterative method improves on both.

**1.5 mrad: bulk closure, edge stall.** Figure 4 shows the 6-panel diagnostic for 1.5 mrad. The story is qualitatively different. The seed Gaussian with  $\sigma_{\text{seed}} \simeq 21$  cm starts from  $\max |r-1| = 64\%$  and an in-window  $\chi^2$  of  $1.42 \times 10^6$ ; the first  $\sim 5$  iterations make dramatic progress, then the iteration enters a slow plateau driven by bins at the closure-window edge  $|z_r| \simeq 83$  cm. After 30 iterations,  $\chi^2$  in the window has dropped by  $\sim 3100\times$  to  $\simeq 453$ , and  $\max |r-1|$  sits at  $\simeq 0.11$ , set by those stuck edge bins. Inside the bulk  $|z_r| < 30$  cm closure is at the  $\sim 1\text{--}2\%$  level on every bin; between 30 and 60 cm the residual climbs to  $\sim 5\text{--}10\%$ . The recovered  $w(z)$  shows a strong central core boost and a clear non-Gaussian shape in the  $|z| \simeq 40\text{--}60$  cm region (truth-vertex axis) that no symmetric Gaussian ansatz can represent.



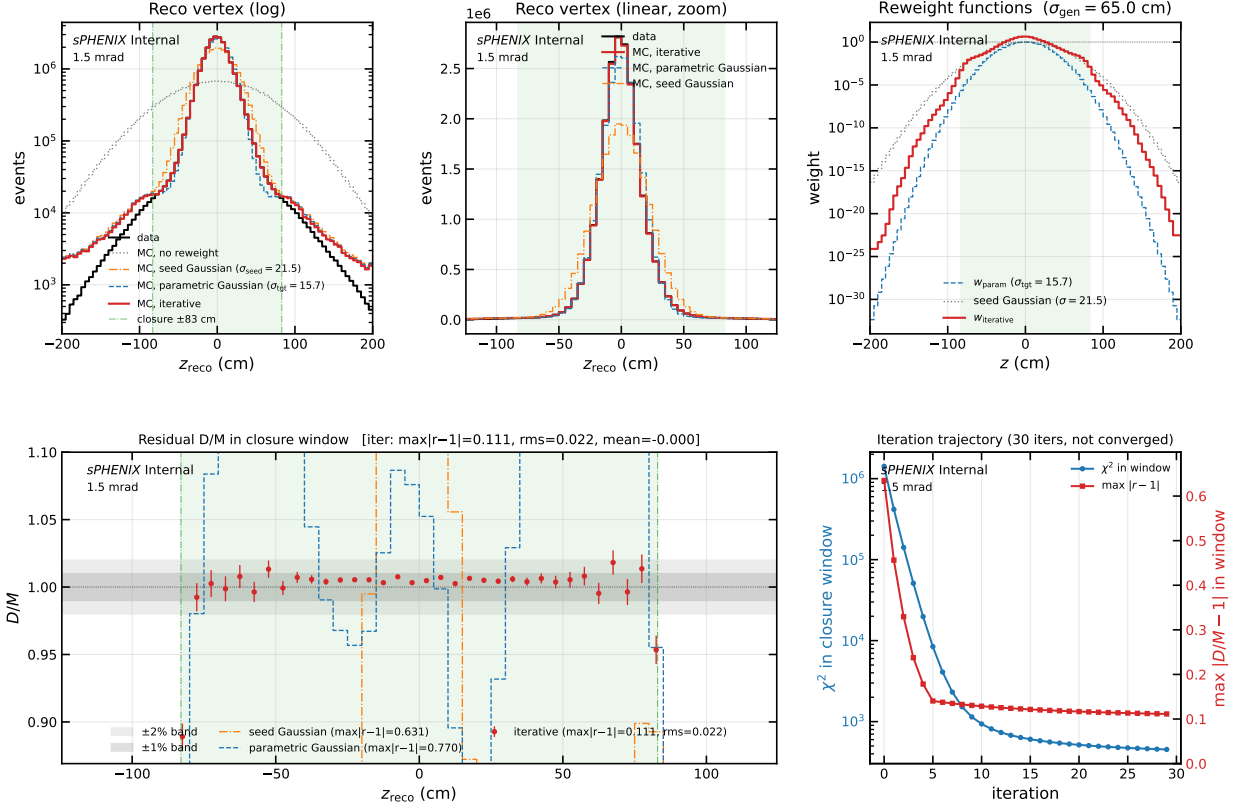


Figure 4: 1.5 mrad iterative closure (6 panels), same layout as Figure 3. At 1.5 mrad the seed Gaussian ( $\sigma_{\text{seed}} = 21.5$  cm, orange dash-dot) and the parametric best-fit Gaussian ( $\sigma_{\text{tgt}} = 15.7$  cm, blue dashed) are distinct; both are much broader than the data core and mis-predict the reco-vertex shape by  $\sim 30$ – $60$  % at the closure-window edge. The iterative method (red points) closes the bulk of the window to  $\sim 1$ – $2$  % inside  $|z_r| < 30$  cm and stalls at  $\max|r-1| \simeq 0.11$  at the closure-window edge  $|z_r| \simeq 83$  cm. The recovered  $w(z)$  in the top-right panel shows a strong central core boost and a clear non-Gaussian shape in the  $|z| \simeq 40$ – $60$  cm region (truth-vertex axis).

**Method comparison.** Figure 5 puts the iterative reweight against three reference baselines at 1.5 mrad: no reweight, the seed Gaussian (the  $\mathcal{N}(0, \sigma_{\text{seed}}^2)$  initial point of the iteration, with  $\sigma_{\text{seed}} = 21.5$  cm measured from  $\text{std}(z_r^{\text{data}})$ ), and the parametric best-fit Gaussian (a single-parameter fit scanning  $\sigma_{\text{tgt}}$  against data, best at  $\sigma_{\text{tgt}} = 15.7$  cm). (Stage 1+2, a parametric residual correction on top of the parametric Gaussian, is never used in the nominal pipeline and is not shown.) The no-reweight MC overshoots data by a factor of several everywhere; both Gaussian baselines close the width on average but mismatch the shape at the 30–70 % level, with visible structure in the ratio panel that no single-Gaussian ansatz can represent; the iterative method is the only one that follows the narrow data core while also tracking the non-Gaussian mid-tail structure.

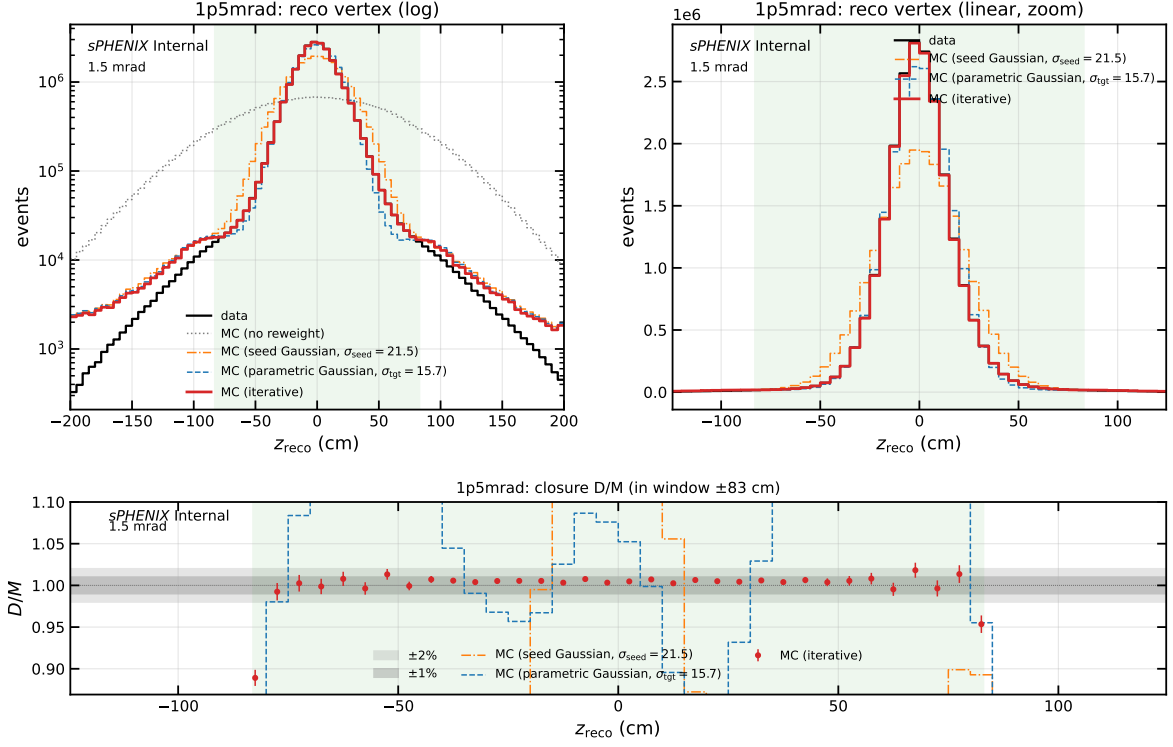


Figure 5: Method comparison at 1.5 mrad: data reco vs mixed MC reco under no reweight (gray dotted), seed Gaussian  $\sigma_{\text{seed}} = 21.5$  cm (orange dash-dot), parametric best-fit Gaussian  $\sigma_{\text{tgt}} = 15.7$  cm (blue dashed), and the iterative data-driven reweight (red), with a ratio panel below. The two Gaussian baselines bracket the “best possible” single-Gaussian description of the data reco shape; neither closes below the 10–30 % level because the true data-MC ratio at 1.5 mrad is non-Gaussian. The iterative method is the only one that follows the narrow data core while also tracking the non-Gaussian mid-tail structure.

**Parametric  $\sigma_{\text{tgt}}$  scan.** Figure 6 shows the stage 1 parametric  $\chi^2$  scan that precedes the iterative method. A single Gaussian truth-vertex weight  $w(z) \propto \mathcal{N}(0, \sigma_{\text{tgt}}^2) / \mathcal{N}(0, \sigma_{\text{gen}}^2)$  is evaluated at each scan point. At 1.5 mrad the minimum is at  $\sigma_{\text{tgt}} \simeq 15.7$  cm with  $\chi^2/\text{ndf} \simeq 3237$ ; the iterative method reaches 13.7, a  $\sim 236\times$  improvement. At 0 mrad the minimum is at  $\sigma_{\text{tgt}} \simeq 54.6$  cm with  $\chi^2/\text{ndf} \simeq 454$ ; the iterative method reaches 2.78, a  $\sim 163\times$  improvement. The gap between the best parametric point and the iterative floor quantifies the shape mismatch that no single-Gaussian ansatz can capture.

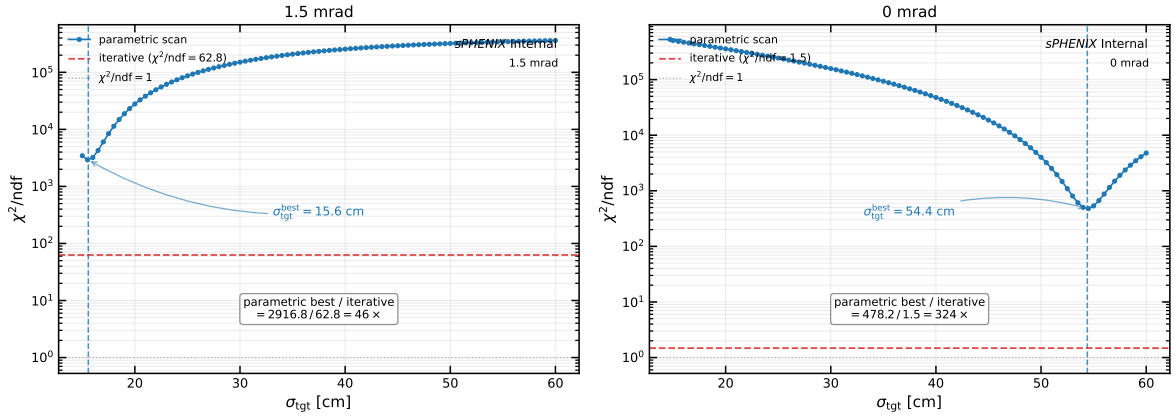


Figure 6: Stage 1 parametric  $\chi^2/\text{ndf}$  as a function of the target truth-vertex Gaussian width  $\sigma_{\text{tgt}}$ , for the 1.5 mrad (left) and 0 mrad (right) periods. The minimum of the scan identifies the best single-Gaussian approximation to the data reco-vertex shape; the parabolic-refined best-fit  $\sigma_{\text{tgt}}$  is marked by a vertical dashed line. The horizontal dashed line (red) shows the final  $\chi^2/\text{ndf}$  achieved by the iterative data-driven method, and the dotted line marks  $\chi^2/\text{ndf} = 1$  for reference. All  $\chi^2$  values are computed on the reco-vertex axis  $z_r$  inside the closure window and normalized by the number of degrees of freedom.

## 5 Physics interpretation

**Why 0 mrad is easy.** At 0 mrad the beams overlap head-on along their full length and the luminous region is broad ( $\sigma_{\text{lum}} \simeq 54 \text{ cm}$ ), nearly equal to the generator beam profile ( $\sigma_{\text{gen}} \simeq 65 \text{ cm}$ ). The shape of the data reco vertex is also close to a single Gaussian. The seed is therefore already a good first approximation, the initial residual is  $\sim 8\%$ , and the iteration just cleans up small departures (slightly enhanced central core). Convergence in a handful of iterations is what one would expect from a near-textbook deconvolution problem.

**Why 1.5 mrad is hard.** Three things conspire at 1.5 mrad:

- The Piwinski crossing-angle effect compresses the luminous region to  $\sigma_{\text{lum}} \simeq 21 \text{ cm}$ , almost a factor of three narrower than the generator profile. The reweight must extract a narrow target from a wide source, and the dynamic range of  $w(z)$  at the central core is correspondingly large ( $w(0) \simeq 8$ ).
- The data reco distribution is *not* a clean Gaussian. It has a narrow core plus a mid-tail shoulder near  $|z_r| \simeq 45\text{--}60 \text{ cm}$ . The iterative method discovers this structure directly — the recovered  $w(z)$  is non-Gaussian in the same  $z$  range — whereas any symmetric one-parameter Gaussian ansatz cannot represent it. This is why the parametric best-fit Gaussian gives  $\chi^2/\text{ndf} \simeq 3237$  on the 1.5 mrad mix, and why the iterative method drops it to  $\simeq 14$  (a further  $\sim 236\times$  improvement).
- The MBD resolution tail leak from core-truth events fills the bins near the closure-window edge ( $|z_r| \simeq 83 \text{ cm}$ ). The iteration becomes insensitive to  $w(z)$  updates at that  $z$  because the contributions to those reco bins come dominantly from truth vertices ( $z_h, z_{\text{mb}}$ ) at other  $z$  values via the wide double-interaction reco-truth kernel. The iteration has the wrong handle on those bins, and that is what sets the residual floor at  $\max|r-1| \simeq 11\%$ .

**What the recovered  $w(z)$  looks like.** At 1.5 mrad,  $w(z)$  has three distinguishing features visible in the top-right panel of Figure 4: (a) a strong central peak with  $w(0) \simeq 8$  that pulls double-

interaction events whose truth vertices ( $z_h$  and  $z_{mb}$ ) both lie close to the IP into the data-like core; (b) a mid- $z$  dip that suppresses the generator’s natural  $|z| \sim 30\text{--}50\text{ cm}$  population to match the narrow data core; and (c) a clear  $\pm$  asymmetry in  $|z| \simeq 40\text{--}60\text{ cm}$  tracking the asymmetric mid-tail bump in the data. None of these features is representable by a symmetric single-parameter Gaussian. At 0 mrad,  $w(z)$  is essentially the seed Gaussian with a small central enhancement.

## 6 Caveats and next steps

**(i) photon10-only mix.** The current fit uses `photon10 + photon10_double` only. The full production blend also includes `jet12 + jet12_double`. Extending the fit to all four samples is the obvious next step: mechanically it is two more `load_sample` calls plus the appropriate cluster-weighted blend, but the closure numbers will move and should be remeasured before any production hookup.

**(ii) 1.5 mrad is not formally converged.** The 30-iteration cap was hit. Two questions remain open: (a) whether extending to  $\sim 100$  iterations would push the stuck edge bin to convergence, or whether it is a hard physical floor set by the double-interaction reco-truth kernel (the suspicion is the latter); and (b) whether the failure to converge in  $\max|r - 1|$  matters for the downstream analysis. The latter has a fairly clean answer: the PPG12 analysis applies fiducial cuts  $|z_r| < 60\text{ cm}$  on the reco vertex and  $|z_h| < 60\text{ cm}$  on the primary truth vertex (Section 2), and inside those cuts the per-bin closure is already at the 1–2 % level. The stuck edge bin at  $|z_r| \simeq 83\text{ cm}$  is well outside the analysis fiducial, so the iteration is good enough for the analysis use case, even though it is not formally converged on the full closure-window metric.

**(iii) Downstream hookup is wired in but off by default.** Both `ShowerShapeCheck.C` and `RecoEffCalculator_TTreeReader.C` now accept two new analysis fields, `truth_vertex_reweight_on` and `truth_vertex_reweight_file`; when `on=1`, the code opens `reweight.root`, reads `h_w_iterative`, and multiplies the per-event weight by  $w(z_h)$  for single-interaction MC and  $w(z_h)w(z_{mb})$  for double-interaction MC. The secondary truth vertex branch `vertexz_truth_mb` (i.e.  $z_{mb}$ ) is picked up through an optional-pointer guard, so single-MC trees without it fall through to the single-vertex path. The two reco-level and truth-level reweights are mutually exclusive and the config loader enforces this by forcing `vertex_reweight_on=0` when `truth_vertex_reweight_on=1`. The showershape configs `config_showershape_{1p5rad,0rad}.yaml` are pre-wired to the matching `output/{1p5mrad,0mrad}/reweight.root`; the nominal BDT configs carry the same fields disabled, so flipping the main analysis over to the truth-vertex reweight is a one-line config edit once the jet-sample extension is done.

**(iv) Implementation can be sped up.** Each iteration currently rebuilds the MC reco histogram from scratch ( $\mathcal{O}(N_{\text{events}})$  per pass). Precomputing  $z^2$  and dropping the per-trial mask rebuild would give a  $\sim 10\times$  speedup with no change in result. Not a blocker; nice-to-have for the all-samples extension.

## 7 Summary

$f(z_r)$  cannot close the individual truth-vertex distributions ( $z_h$  and  $z_{mb}$ ) of double-interaction MC, which biases the primary truth fiducial cut  $|z_h| < 60\text{ cm}$  used in the efficiency denominator by +1.4 % at 1.5 mrad (Section 2). An iterative data-driven reweight  $w(z)$ , applied per truth vertex with the factorization  $w(z_h)w(z_{mb})$  for double-interaction events, closes this gap without any Gaussian assumption. Applied to the photon10 mix, the method converges cleanly at 0 mrad ( $\max|r - 1| < 1\%$  in 8 iterations) and reduces in-window  $\chi^2$  by a factor  $\sim 3100$  at 1.5 mrad, closing the analysis-relevant bulk  $|z_r| < 30\text{ cm}$  to 1–2 % per bin and reaching  $\sim 11\%$  at the closure-window

edge. The recovered  $w(z)$  at 1.5 mrad shows non-Gaussian features (a strong central boost and a mid- $z$  shoulder) that no closed-form Gaussian ansatz could represent. Per-period `reweight.root` outputs are ready in `efficiencytool/truth_vertex_reweight/output/` and the C++ hookup into `ShowerShapeCheck.C` and `RecoEffCalculator_TTreeReader.C` is in place (enabled on the showershape period configs, wired but off on the nominal BDT configs). The remaining work is the jet-sample extension of the standalone fit tool.

## A Baseline: events with and without a reconstructed vertex

Before building any truth-based reweight, we checked that dropping events whose reco MBD vertex is the  $-9999$  sentinel does not itself bias the MC. The concern is that an MC event which loses its reco vertex  $z_r$  may be physically unusual (e.g. with a shifted truth vertex  $z_h$ ), so dropping it would skew the efficiency in a  $z_h$ -dependent way.

| Sample          | $N_{\text{total}}$ | $N_{\text{reco}}$ | $N_{\text{no reco}}$ | fraction no reco |
|-----------------|--------------------|-------------------|----------------------|------------------|
| photon10        | 9 998 562          | 7 716 585         | 2 281 977            | <b>22.82 %</b>   |
| photon10_double | 9 997 075          | 9 491 223         | 505 852              | <b>5.06 %</b>    |
| jet12           | 10 000 000         | 7 689 530         | 2 310 470            | <b>23.10 %</b>   |
| jet12_double    | 9 997 000          | 9 485 384         | 511 616              | <b>5.12 %</b>    |

Table 2: Fraction of events without an MBD reco vertex. The double-interaction productions have  $\sim 4.5\times$  higher MBD vertex-finding efficiency than their single-interaction counterparts (two concurrent collisions give the MBD twice the chance of producing a vertex).

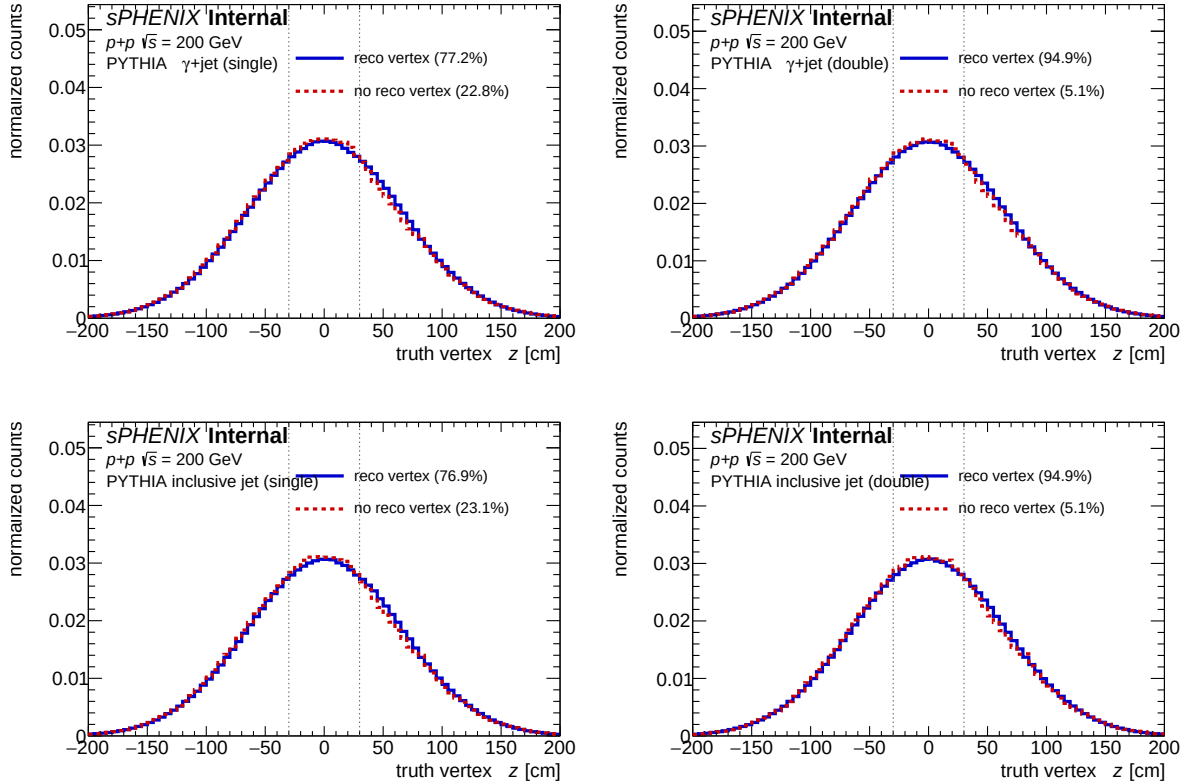


Figure 7: Primary truth vertex  $z_h$  distribution for events with (blue) and without (red dashed) a reconstructed MBD vertex  $z_r$ , area-normalized for shape comparison. Columns: single / double interaction. Rows: photon10 / jet12. Dotted vertical lines:  $|z_h| = 30$  cm reference.

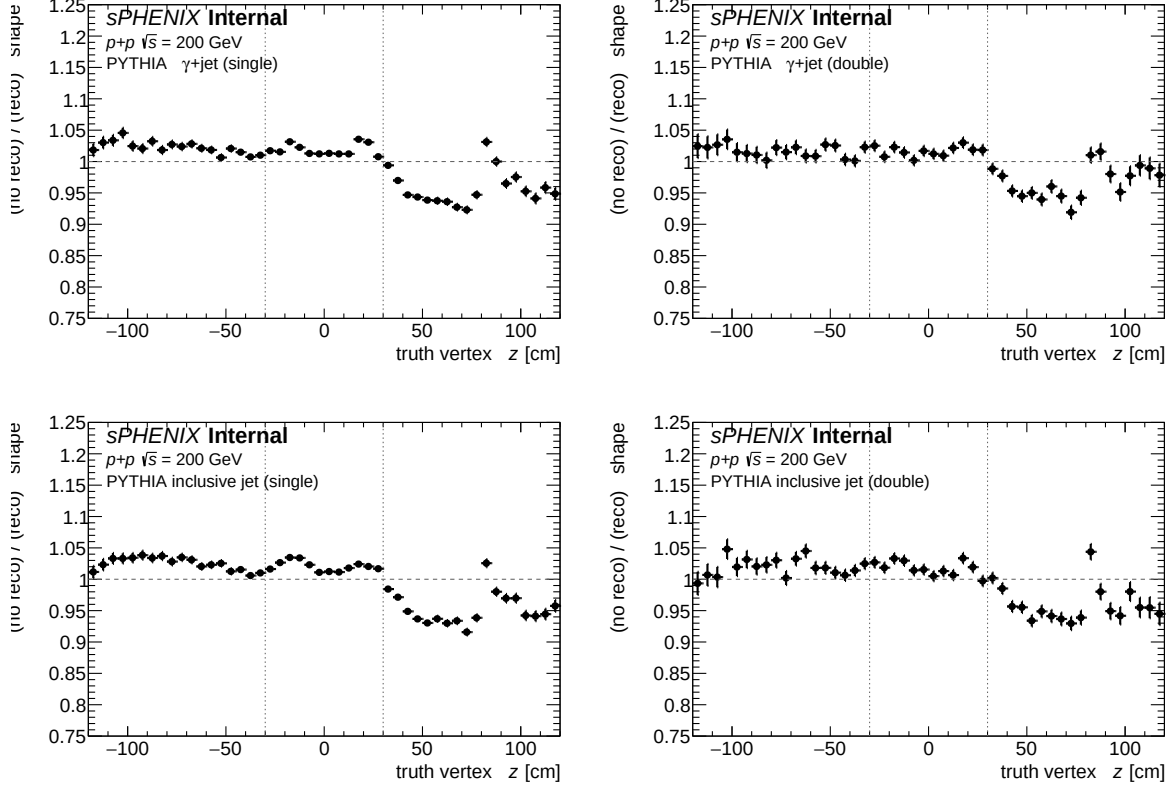


Figure 8: Ratio (no-reco-vertex shape) / (reco-vertex shape) as a function of primary truth vertex  $z_h$ , zoomed to  $[0.75, 1.25]$ . All samples show a  $\sim \pm 5\%$  north/south asymmetry inside  $|z_h| < 30$  cm, consistent with a small MBD geometric bias.

Inside  $|z_h| < 30$  cm the two shapes agree to within  $\pm 5\%$  with a small  $z_h$ -asymmetry that is the same in all four samples and therefore a detector-level effect (MBD vertex finder slightly more efficient at positive  $z$ ). Dropping the no-reco-vertex events is thus *approximately* harmless — both the efficiency numerator and denominator are scaled by the same factor, and the residual  $z$  dependence of the drop rate is small. However, dropping  $\sim 22\%$  of the single-interaction MC sample is still a lot of statistics, and it makes the efficiency measurement implicitly conditional on a valid reco vertex being found. That was the original motivation for exploring a truth-level reweight in the first place.

## B Discarded methods

Before arriving at the iterative method described in Section 3, we explored three truth-only reweight constructions that did not work:

- **$w_1$ : direct ratio** of the data reco-vertex shape to the MC truth-vertex shape,  $w_1(z_h) = D(z_r)/T(z_h)$ , with the common binning identifying  $z_r$  bins with  $z_h$  bins.
- **$w_2$ : conditional projection**,  $w_2(z_h) = \mathbb{E}[f(z_r) | z_h]$ , the expectation of the production reco weight conditional on the primary truth vertex.
- **$w_{\text{iter,DA}}$ : D’Agostini iterative unfolding** using the MC response matrix  $R_{ij} = P(z_r^i | z_h^j)$ .

All three hit a geometric lower bound on the double-MC sample at 1.5 mrad: the conditional reco kernel  $P(z_r | z_h)$  (for single-interaction MC) has RMS  $\sim 34$  cm, wider than the  $\sim 22$  cm target data peak (Section C), so *any* truth-only reweight is bounded from below by the kernel width and cannot reach the target.

We also drafted a closed-form **2D Gaussian reweight**  $w(z_h, z_{\text{mb}}) \propto \exp[-\frac{1}{2}(z_h^2 + z_{\text{mb}}^2) \Delta]$  with  $\Delta = 1/\sigma_{\text{lum}}^2 - 1/\sigma_{\text{MC}}^2$ . This closed form does work in principle — it acts on both the symmetric and antisymmetric coordinates and so can narrow all three vertex axes simultaneously — but it assumes a single Gaussian shape for the data luminous region. At 1.5 mrad the data has a  $\pm$ -asymmetric mid-tail bump that a symmetric Gaussian cannot reproduce, so the closed form ended up as a first-draft proposal that was superseded by the iterative method of Section 3, which discovers  $w(z)$  from data directly without any Gaussian assumption.

## C Double-interaction reco-truth kernel

The geometric lower bound on truth-only reweights quoted in Section B can be read off the widths directly:

- 1.5 mrad data reco width:  $\sigma \simeq 22.3$  cm
- jet12 single-interaction MC truth width ( $z_h$ ):  $\sigma \simeq 57.6$  cm
- jet12\_double MC truth width ( $z_h$ , primary):  $\sigma \simeq 62.7$  cm
- jet12\_double MC reco width ( $z_r$ ):  $\sigma \simeq 48.6 \approx 62.7/\sqrt{2}$  (averaging)
- Conditional kernel  $P(z_r|z_h=0)$  RMS (with  $z_{\text{mb}}$  marginalized):  $\sim 34$  cm

Figure 9 shows the kernel slice directly, compared to the 1.5 mrad data target. Even if the primary truth vertex is forced to  $z_h = 0$ , the reco distribution from a double-interaction event cannot be narrower than the  $\sim 34$  cm kernel width — larger than the 22 cm target. This is the structural reason the truth-only methods fail on double-interaction MC, and the structural reason the iterative method needs the  $w(z_h)w(z_{\text{mb}})$  factorization rather than a  $w$ -on- $z_r$  construction.

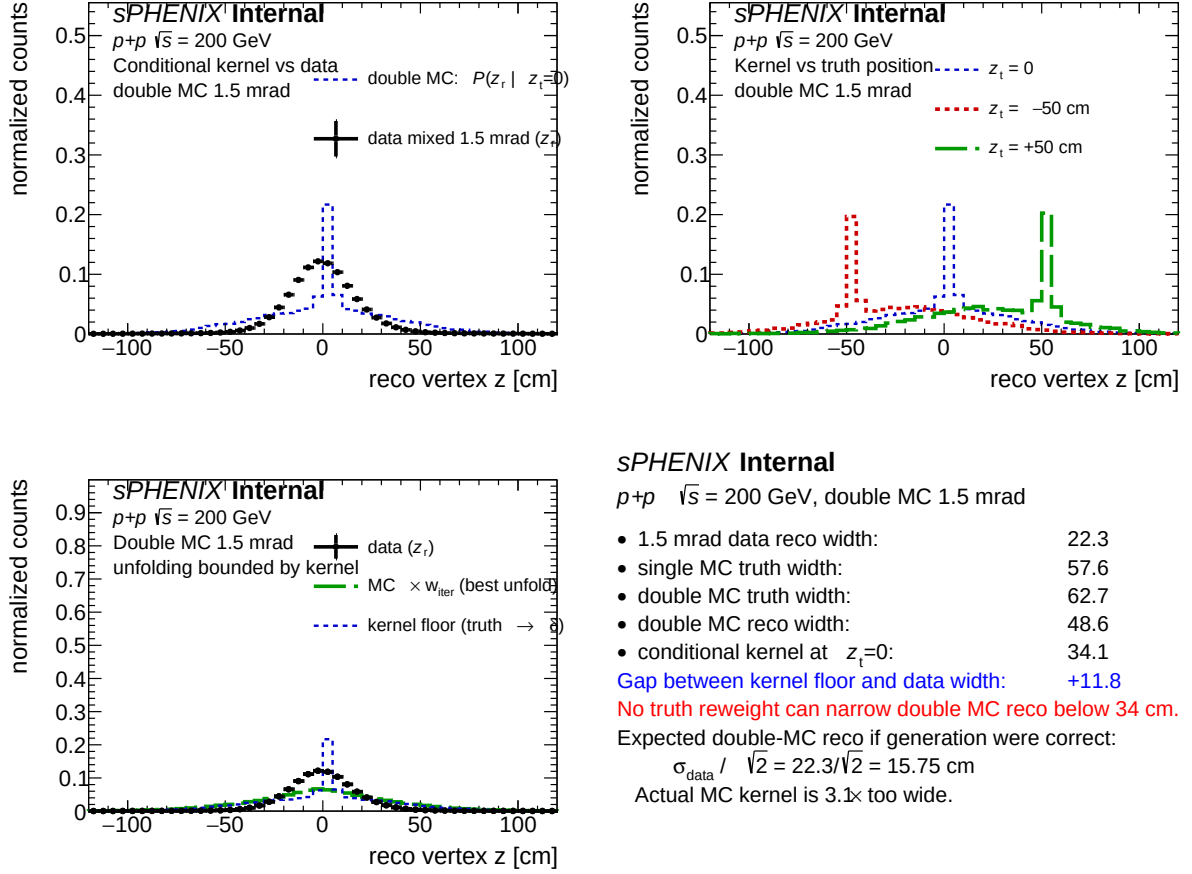


Figure 9: Conditional reco kernel  $P(z_r|z_h)$  in jet12\_double MC at 1.5 mrad (with  $z_{\text{mb}}$  marginalized), with the 1.5 mrad data reco  $z_r$  shape overlaid. The kernel's  $\sim 34$  cm RMS exceeds the  $\sim 22$  cm data peak, which is the geometric lower bound on truth-only reweighting.

## D Files and code

- `anatreemaker/source/CaloAna24.{h,cc}` — writes the `vertexz_truth` ( $z_h$ , primary truth vertex) and `vertexz_truth_second` ( $z_{\text{mb}}$ , secondary truth vertex; read by C++ as `vertexz_truth_mb`) branches that the iterative method reads for single- and double-interaction MC respectively.
- `efficiencytool/truth_vertex_reweight/` — the standalone Python tool for the iterative method:
  - `fit_core.py` — cached sample loader, the `mixed_mc_histogram` forward model (Eq. 2), Poisson  $\chi^2$ , and helpers for fitting  $\sigma_{\text{gen}}$  from MC truth.
  - `fit_truth_vertex_reweight.py` — the driver (stage iterative = Section 3), producing `reweight.root` and `fit_result.yaml`.
  - `plot_closure.py`, `plot_comparisons.py` — closure and cross-period diagnostic plots.
  - `config.yaml` — sample paths, run ranges, fit parameters.
- `efficiencytool/TruthVertexRecoCheck.C` — Appendix A supporting macro.
- Outputs: `efficiencytool/truth_vertex_reweight/output/{1p5mrad,0mrad}/reweight.root`, `fit_result.yaml`, and the cached event arrays in `output/cache/`.